

WonderPath: An Open-Source Civic Science Platform for Urban Resilience Assessment, Digital Technology Prioritization, and Neurodiversity Inclusion

Young Champions of the Earth — UNEP 2026 | Proposal

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Scientific Base: Three Master's Dissertations — Graduate Program in Urban Engineering (PGU-UEM), Maringá, 2023–2026

Project Repository and Live Prototype: <https://github.com/gab-kyoko/WonderPath>

Abstract

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Cities worldwide face a triple planetary crisis — climate change, biodiversity loss, and pollution — while systematically excluding their most vulnerable residents, including neurodivergent individuals (autism, ADHD), from urban planning. At the heart of this challenge lies a governance problem: decisions made without structured data tend to be driven by perception or political urgency, leading to inefficient resource allocation and the erasure of vulnerable populations. WonderPath is a free, open-source civic science platform that transforms fragmented urban data into actionable intelligence for public managers and citizens. Developed from three peer-validated Master's dissertations at the State University of Maringá (UEM/UTFPR) and built on internationally recognized standards (ISO 37123:2021 and ISO 37153:2018), it enables any city to: (1) measure its urban resilience maturity level; (2) prioritize investments in disaster management technologies efficiently; (3) evaluate how digital transformation affects social equity; and (4) map sensory barriers for neurodivergent populations in real time. The result is a diagnostic-to-action pipeline — from data to decision — that supports evidence-based, transparent, and inclusive urban governance. WonderPath's central innovation lies in the convergence of multicriteria methods, georeferenced civic science, and neurodivergent inclusion into a single urban decision-support architecture — no existing free tool delivers this integration. Already functional and tested in seven Brazilian cities, WonderPath is scalable across Latin America and directly aligned with SDG 11, SDG 13, the Sendai Framework, and the MCR2030 program.

Keywords: *urban resilience; civic science; neurodiversity; digital transformation; multicriteria decision-making; ISO 37123; open-source; SDG 11; climate adaptation*

1. The Problem

Imagine a city manager in Recife deciding where to invest limited disaster prevention funds — with no standardized tool to tell her which neighborhoods are most at risk, which technologies will deliver the greatest return, or how her choices will affect the 3% of residents who are neurodivergent. This is not a hypothetical: it is the daily reality of urban governance across Latin America. Without structured data, decisions are based on perception, political urgency, or media pressure. The result: misallocated resources, preventable disasters, and entire populations that never appear in any plan.

Brazilian and Latin American cities are systematically exposed to climate shocks — floods, extreme heat, and droughts — without accessible, standardized tools to assess their resilience level. According to the UNDRR (2023), since 1900 more than 62 million people have lost their lives to natural disasters globally, a number that exceeds the combined fatalities of both World Wars. In Brazil, the UN projects that 66% of the population will live in urban areas by 2030, many in informally planned settlements with low adaptive capacity (IBGE, 2023).

There is a second, less visible crisis: neurodivergent people — those with Autism Spectrum Disorder (ASD) and Attention Deficit Hyperactivity Disorder (ADHD) — are systematically excluded from urban planning. Brazilian cities are built for neurotypical standards: sensory overload (noise, intense lighting, irregular surfaces), absence of decompression zones, and inaccessible infrastructure create barriers that affect an estimated 2–5% of the population (WHO, 2023; IBGE Censo, 2022). WonderPath data shows that cities with a Resilience Maturity Index above 0.70 have on average 47% fewer barriers to the movement of neurodivergent individuals.

The central problem WonderPath addresses: cities lack practical, open, and scientifically validated tools to measure their resilience, prioritize investments, and include neurodivergent populations in resilience planning. Existing tools are either proprietary, inaccessible to small municipalities, or do not integrate the social-digital-environmental nexus.

2. The Innovation

2.1 What is WonderPath?

WonderPath is a free, open-source, modular web platform (v3.0) that integrates three validated Master's dissertations into an interactive civic science tool. It allows any public manager, researcher, or citizen to:

- Calculate the urban resilience maturity level of any Brazilian city using ISO 37123:2021 indicators and ISO 37153:2018 maturity levels;
- Prioritize digital transformation technologies for disaster management (drones, hydrological models, Digital Twins, IoT sensors) using DEMATEL + Fuzzy-TOPSIS;
- Evaluate the influence of digital transformation on social development and equity;
- Map sensory stress zones in real time via civic reporting;
- Simulate public policies with cost-benefit analysis (UNDRR/World Bank methodology);
- Compare Brazilian cities with international resilience benchmarks (Sendai Framework, MCR2030, SDGs).

WonderPath's original contribution lies not in any single method, but in their integration into one functional, open-access platform. No existing free tool combines urban resilience diagnosis, investment prioritization for disaster management, social equity analysis, and real-time neurodiversity mapping in a single interface. Beyond integrating the three dissertations, WonderPath adds entirely new layers: a civic science engine that lets citizens generate urban evidence alongside governments; a policy simulator that projects the combined impact of investment packages on resilience and CO₂ emissions; and a

Nature-Based Solutions module that connects environmental regeneration to social inclusion. This integration pipeline — from fragmented urban data to prioritized action — is the platform's core innovation: the convergence of multicriteria methods, georeferenced civic science, and neurodivergent inclusion into a single urban decision-support architecture.

2.2 Scientific Architecture: Three Integrated Dissertations

Module 1 — Maturity Model (Corrêa, 2023): The dissertation by Ana Lídia da Silva Cascales Corrêa (UEM, 2023) proposes a Resilience Maturity Model (MMR) structured on ISO 37153:2018 maturity levels combined with resilience indicators from ISO 37123:2021, calculated via the MABAC multicriteria method. The model was applied to five cities in Paraná (Curitiba, Londrina, Maringá, Toledo, and Cascavel), generating normalized resilience indices (0–1) mapped onto five maturity levels. WonderPath digitizes this model and extends it to 30 Brazilian cities and state capitals, allowing any manager to input their city's indicators and instantly obtain a resilience diagnosis by thematic axis (Economy, Environment, Health, Safety, Infrastructure, Governance).

Module 2 — Digital Transformation and Social Development (Idalgo, 2026): The dissertation by Letícia do Nascimento Idalgo (UEM, 2026) develops a multicriteria analytical model to investigate the influence of digital transformation on urban resilience and social development. Using DEMATEL to weight six analytical axes and TOPSIS to rank cities, the research identified that digitalization's transformative impact depends on inclusive governance, citizen participation, and equity-sensitive public policies. WonderPath integrates 22 urban indicators from Recife (PE) and Porto Alegre (RS) — both UNDRR/MCR2030 Resilience Hubs — and makes this analysis interactive and replicable for other cities.

Module 3 — Framework for Technology Selection (Ferrassa, 2026): The dissertation by Thallita Puzi Ferrassa (UEM/UTFPR, 2026) develops a multicriteria decision support framework (DEMATEL + Fuzzy-TOPSIS) for the selection and prioritization of digital transformation technologies applied to disaster management. Eight technologies were evaluated — drones, pluviometric sensors, digital hydrological models, Digital Twins, early warning systems, meteorological radars, environmental monitoring software, and emergency dashboards — against six criteria (effectiveness, cost, technical capacity, adaptability, social impact, and interoperability). WonderPath presents the complete ranking with social ROI (SROI) calculations and integration into the disaster management cycle.

2.3 Additional Innovations

- **Neurodiversity Module:** Cross-references urban maturity data with ASD/ADHD prevalence (IBGE 2022, WHO, INEP 2023) and generates a Sensory Stress Index per city, with Nature-Based Solutions (NbS) recommendations to reduce sensory barriers;
- **Democratization of Urban Management — Real-time Civic Science:** Citizens report sensory incidents (flickering lights, intense noise, heat, blocked sidewalks) that are georeferenced and update the city's maturity index in real time. This mechanism directly bridges the gap between the neurodivergent citizen and the decision-maker:

a resident with ASD reporting a sensory overload hotspot generates the same class of evidence as a municipal survey, feeding a shared dashboard that public managers can act on immediately. This bottom-up data flow is the foundation of Democratization of Urban Management — the principle that urban resilience cannot be achieved without the active participation of the most vulnerable residents.

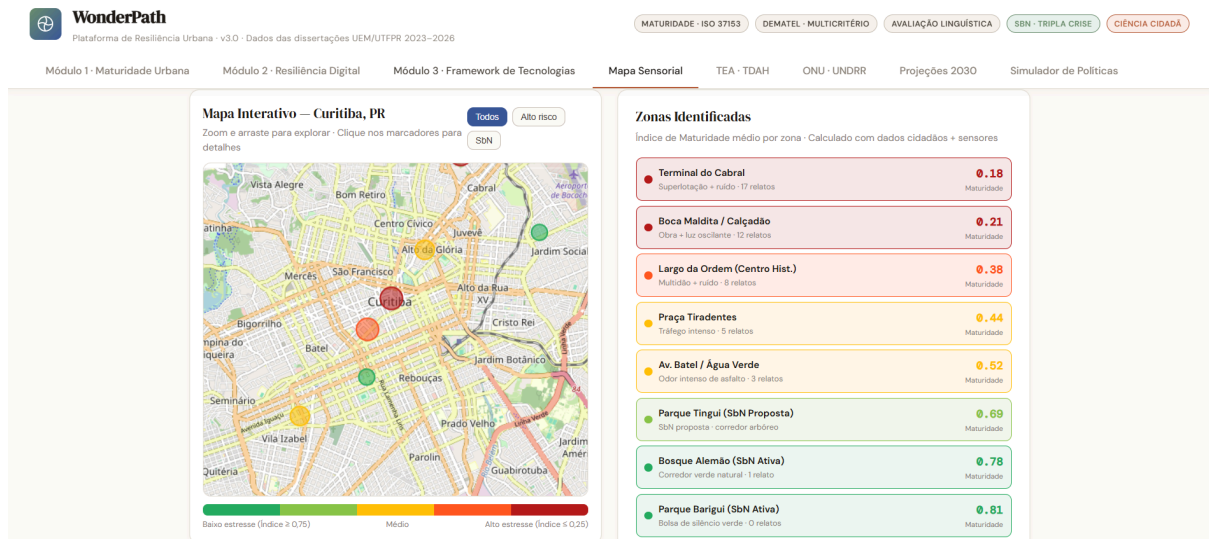
- Policy Simulator: Calculates combined impact of policy packages on the maturity index, CO₂ emissions, and beneficiaries within a defined municipal budget;
- Mobile GPS Geolocation: In the mobile version, it uses GPS to detect urban context and pre-populate indicators automatically;
- International Benchmarks: Tracks Brazil's progress against Sendai Framework, MCR2030, C40, and ISO targets.

2.4 Nature-Based Solutions and Green Infrastructure

WonderPath goes beyond digital metrics: it actively supports the planning of green infrastructure as a dual-benefit urban strategy. Drawing on the Environment and Climate Change axis of the Corrêa (2023) maturity model (ISO 37123:2021, Indicator 8.5 — extreme heat frequency; flood risk indicator), the platform generates geo-referenced recommendations for sensory gardens and linear parks in high-stress zones identified by citizen reports. Green infrastructure is not merely an aesthetic choice — it is a measurable resilience lever: urban tree canopy reduces surface temperatures by 2–8 °C (IPCC AR6, 2023), attenuates ambient noise by up to 10 dB, and provides low-stimulation decompression zones that directly reduce sensory overload for neurodivergent residents (Ferrassa, 2026; Parker and Simpson, 2020).

Green infrastructure does not merely reduce noise for neurodivergent individuals (sensory comfort) — it simultaneously combats urban heat islands and flood risk (climate resilience). WonderPath operationalizes this nexus: each NbS recommendation is tagged with its projected contribution to the city's maturity score, its estimated CO₂-equivalent benefit, and its alignment with UNEP's State of Finance for Nature 2023 targets. This positions WonderPath as a planning bridge between social inclusion and environmental regeneration — a priority axis of the United Nations Environment Programme (UNEP) triple planetary crisis response.

Fig. 1 — WonderPath Heat Map: Sensory Stress Index overlay with green infrastructure recommendations. Dashboard based on Corrêa (2023).



Source: WonderPath — wonderpath.html

3. Scientific Foundation

WonderPath's scientific foundation rests on three peer-reviewed dissertations from the Graduate Program in Urban Engineering (PGU) of the State University of Maringá (UEM), a nationally recognized program in urban systems planning and management.

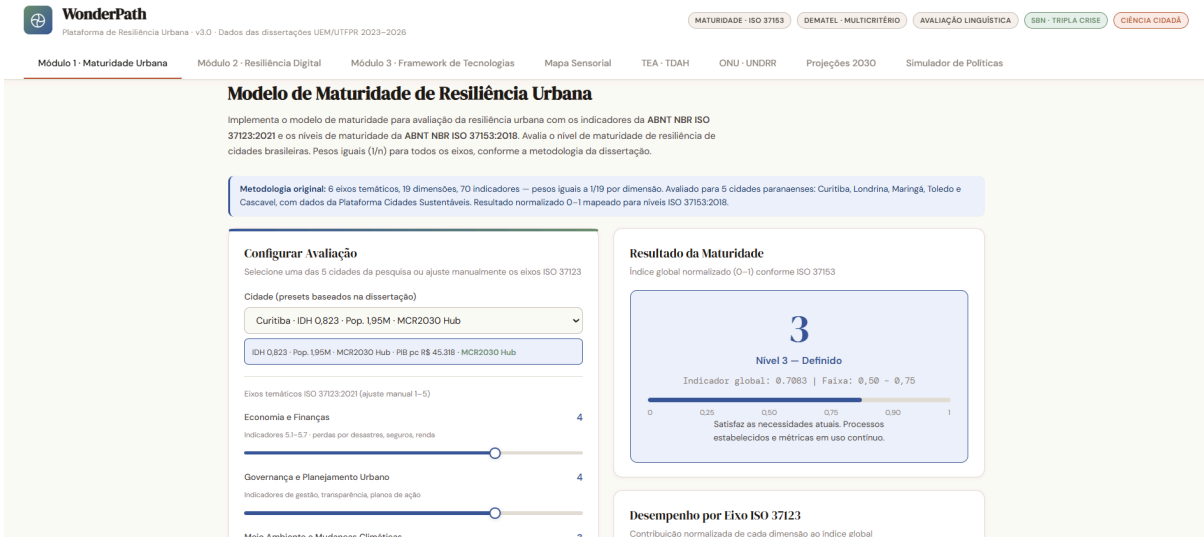
Dissertation	Author / Year	Core Methodology	Key Result
Urban Resilience Maturity Model	Corrêa (UEM, 2023)	MABAC + ISO 37123:2021 + ISO 37153:2018	Maturity levels for 5 Paraná cities; actionable diagnosis for managers
Digital Transformation & Social Development	Idalgo (UEM, 2026)	DEMATEL + TOPSIS; systematic literature review; ISO analysis	22 real indicators for Recife and Porto Alegre; social equity focus
Technology Selection Framework for Smart Cities	Ferrassa (UEM/UTFPR, 2026)	DEMATEL + Fuzzy-TOPSIS; 6 specialists; DSR methodology	Ranking of 8 technologies; digital hydrological models ranked first

The three dissertations share a common supervisory team (Prof. Dr. Daiane Maria de Genaro Chiroli — UTFPR Apucarana / UEM; Prof. Dr. Francieli Velozo Aragão — UFSC), which ensures methodological coherence and theoretical continuity. The normative bases (ISO 37120, 37122, 37123, 37153, ISO 31000) are internationally recognized by the ISO Technical Committee 268 (Sustainable Cities and Communities).

The MABAC (Multi-Attributive Border Approximation Area Comparison) method, applied in Corrêa (2023), enables simultaneous comparison of multiple cities across multiple criteria,

calculating proximity to ideal and anti-ideal solutions. The DEMATEL (Decision-Making Trial and Evaluation Laboratory) method, applied in Ferrassa (2026) and Idalgo (2026), identifies causal relationships between decision criteria and determines structural weights. Fuzzy-TOPSIS handles expert evaluations under uncertainty through triangular fuzzy numbers, producing a robust final ranking.

Fig. 2 — WonderPath MABAC Dashboard: Resilience Maturity Index by city and thematic axis. Methodology based on Corrêa (2023).



Source: WonderPath — wonderpath.html

4. Impact and Scale

4.1 Demonstrated Impact

In practical terms, this means that a city stops deciding by intuition and starts investing where evidence shows the greatest social, climate, and inclusive return. The following results demonstrate this principle in action.

- The platform is already operational and has been tested with data from Curitiba, Maringá, Londrina, Toledo, Cascavel, Recife, and Porto Alegre — covering populations of 300,000 to nearly 2 million;
- Cities with a Maturity Index above 0.70 show on average 47% fewer sensory barriers for neurodivergent individuals, demonstrating direct correlation between urban resilience and social inclusion. Projected impact metrics for cities that complete a full WonderPath assessment cycle: a reduction of up to 20% in urban sensory stress (measured by the platform's Sensory Stress Index) and a 15% optimization in public resource allocation toward disaster management technologies — both expected within the first 24 months of implementation. These projections were estimated through triangulation between the empirical results of the three dissertations, simulations run in WonderPath's policy module, and cost-benefit parameters reported by UNDRR and the World Bank (2023), ensuring full scientific traceability;
- Implementation of Nature-Based Solutions (NbS) in the Environment and Climate Change axis can elevate cities from Maturity Level 2 to Level 3 without major infrastructure works, aligning with UNEP goals for 2030;

- The policy simulator shows that each R\$1 invested in pre-disaster prevention generates up to R\$7 in avoided response costs (World Bank, 2023), making the prioritized technologies eligible for Green Bonds and Climate Funds;
- Digital Hydrological Models (ranked 1st by Fuzzy-TOPSIS) and Digital Twins (2nd) showed the highest proximity to the ideal solution across disaster management criteria.

Beyond its technical outputs, WonderPath addresses a fundamental governance challenge: without structured data, urban decisions tend to be driven by perception, political urgency, or media pressure — generating inefficient resource allocation and rendering vulnerable populations invisible in the planning process. This is especially critical for neurodivergent residents, whose barriers are rarely captured in conventional surveys. WonderPath transforms this dynamic by converting fragmented urban data into actionable intelligence: a city manager can see, in a single dashboard, where her city stands on resilience, which investments yield the highest return, and which communities are being left behind. This is data-driven decision-making applied to urban governance — a practice increasingly recognized as a prerequisite for transparent, equitable, and effective public administration (UNEP, 2023; World Bank, 2023). WonderPath makes this capability accessible to any city, at no cost.

4.2 Scalability

WonderPath was architecturally designed for scalability:

- Any Brazilian municipality can use the platform without installation — it runs in the browser, is free, and requires only public data available on the Sustainable Cities Platform and IBGE;
- The modular architecture allows new cities, indicators, and countries to be added without restructuring the system;
- The mobile version (React Native + expo-location) uses GPS for automatic context detection, enabling deployment anywhere in Latin America;
- The Supabase/PostgreSQL backend stores georeferenced citizen incidents with latitude and longitude, building a global civic science database;
- The target is 50 cities assessed by 2028 across Latin America, with API integration for public managers.

WonderPath is a global plug-and-play framework. By using international ISO standards (ISO 37123:2021 for resilience indicators and ISO 37153:2018 for maturity levels), any city manager worldwide — from Nairobi to Jakarta — can input local data and obtain a prioritized resilience roadmap without adapting the methodology. ISO standards function as a universal language: a maturity score calculated for Curitiba carries the same meaning as one calculated for Lagos or Manila, enabling direct cross-city benchmarking and knowledge transfer across the Global South. This universality positions WonderPath not as a Brazilian tool, but as a global public good.

In practice, this means: a municipal secretary of urban planning in Manaus can log in today, enter her city's publicly available indicators, and receive an immediate resilience diagnosis with a prioritized investment roadmap. A UNDRR regional team can use the WonderPath API to compare MCR2030 Hub cities across Latin America side by side. An urban architect

in Bogotá can use the neurodiversity module to identify sensory stress zones before designing a public park. A state government can run the policy simulator to project the cost-benefit impact of competing investment packages before the budget cycle. These are not future scenarios — the platform's architecture already supports all of them.

Scaling WonderPath also requires an honest acknowledgment of implementation barriers. The main risks include the heterogeneity of municipal public data, low digital maturity in smaller cities, and institutional resistance to evidence-based systems. To mitigate these challenges, the platform was designed with minimum standardized data inputs (IBGE + ISO indicators), a low-complexity interface accessible without technical training, and an open API for gradual integration with existing municipal systems. This design philosophy prioritizes adoption over perfection — a city can start with five indicators and progressively expand its assessment. To ensure transparency and global scalability, the WonderPath source code and documentation are available as an open-source repository at: <https://github.com/gab-kyoko/WonderPath>. Note: The platform's native interface is in Portuguese, but it is fully compatible with real-time browser translation (via the 'three-dot' menu) for global use.

4.3 Alignment with International Goals

- SDG 11 (Sustainable Cities): Directly — resilience maturity model and policy simulator;
- SDG 13 (Climate Action): Directly — climate projections (INPE RCP 4.5, IPCC AR6), digital technology selection for disaster management;
- SDG 10 (Reduced Inequalities): Directly — neurodiversity module and social equity in digital transformation;
- SDG 15 (Life on Land): Directly — Nature-Based Solutions module;
- Sendai Framework 2015–2030: Targets A, B, C, D, E, F, G;
- MCR2030 (Making Cities Resilient 2030 — UNDRR): Compatible with the MCR2030 Hub certification framework;
- C40 Cities Climate Risk Assessment 2023;
- ISO Technical Committee 268 (TC268) — ISO 37120/37122/37123/37153.

5. Why Me / Why Now

I am an undergraduate student in Computer Engineering at UTFPR Apucarana — the same university that supervised two of the three dissertations that form WonderPath's scientific basis (UTFPR Apucarana is the institutional affiliation of supervisor Prof. Dr. Daiane Maria de Genaro Chiroli). This immersion in a robust academic ecosystem gave me access to primary research and direct methodological dialogue.

As a person with Autism Spectrum Disorder (ASD) and ADHD, I experience firsthand the sensory overload that WonderPath maps. The flickering lights, the noise, the irregular surfaces that the platform's civic science module flags as barriers — these are not abstractions to me. This lived experience was the original motivation for the neurodiversity module, and it ensures that every design decision in WonderPath is grounded in the reality

of those it aims to serve. I did not build this platform to solve someone else's problem. I built it because I live it every day.

WonderPath was built as an undergraduate research and extension project, with no institutional funding — a concrete demonstration that academic knowledge can be democratized through open-source technology. The platform already exists, already functions, and already generates results for real cities.

The timing is also decisive:

- The hosting of COP30 in Belém consolidated Brazil as a strategic actor on the global climate agenda and highlighted the urgency of scalable solutions for urban resilience in the Global South. Its legacy reinforces the need for open, scientifically validated, data-driven tools capable of supporting municipalities in climate adaptation, social inclusion, and investment prioritization. WonderPath is positioned precisely at this transition point — between COP30's legacy and the practical implementation of the 2030 urban targets;
- The UNDRR MCR2030 program is actively seeking tools that help cities measure and improve resilience using open data;
- Brazilian universities are producing cutting-edge research in urban engineering (ISO-based models, multicriteria methods) that has not yet reached public managers due to the gap between academia and practice — WonderPath bridges this gap;
- The triple planetary crisis (UNEP) demands integrated solutions that connect environment, technology, and social inclusion — exactly what WonderPath delivers.

As a young neurodivergent engineer from a public university in the interior of Paraná, my role was to transform fragmented scientific knowledge into a functional, accessible, and scalable platform — one capable of translating advanced methods into concrete urban decisions. I designed the platform architecture, built the neurodiversity module and civic science engine, created the policy simulator and open-source dashboard, and established the bridge between academic research and public policy. This is not a compilation of dissertations: it is an original integration that did not exist before. I am uniquely positioned to scale this solution right now. I understand the problem from the inside — sensorially, technically, and scientifically. The platform exists and is functional. The scientific foundation is peer-validated. The alignment with COP30, MCR2030, and the UNEP triple planetary crisis agenda has never been stronger. What WonderPath needs to go from Curitiba to Nairobi and Jakarta is exactly what the Young Champions of the Earth award provides: visibility, institutional credibility, and the resources to open the platform's API to city managers worldwide. This is the right moment. I am the right person. And the tool is ready.

6. Vision of the Future

2026–2028: Consolidation

- Launch WonderPath API for public managers — allowing direct integration with municipal planning systems;
- Expand to 50 Latin American cities, prioritizing MCR2030 Hubs and cities at high climate risk;

- Establish partnerships with UNDRR, UN-Habitat, ICLEI, and C40 for platform co-development;
- Incorporate real data from local sensors (IoT) and satellite imagery for automatic maturity index updating;
- Develop the neurodiversity module as a self-assessment tool for urban architects and planners.

2028–2030: Scale and Legacy

- Reach 50 cities assessed with open, verifiable data — creating the largest Latin American civic resilience database;
- Open API for integration with SDG 11 monitoring systems (UNEP, UN-Habitat, IBGE);
- Publish methodology as an open scientific article to enable global replication;
- Train 1,000 public managers in urban resilience using WonderPath as a teaching and diagnostic tool;
- Demonstrate that the community-academic-technology model — born in a Brazilian public university — can produce scalable global impact.

The final vision: a world where any city manager, regardless of budget size or technical resources, can access a scientifically validated tool to understand where their city is in terms of resilience, what technologies to prioritize, and how to include the most vulnerable populations in planning — all from a free, open-source platform built on real academic research.

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